

AI4AI4CELL: COLLABORATIVE EVIDENCE-GUIDED RESEARCH FOR BIOLOGICAL MODEL IMPROVEMENT

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ABSTRACT

Biological data are often distributed across institutions and data holders, where privacy and data-sharing constraints limit centralized access. A key challenge for self-improving AI in this setting is to use such distributed evidence to improve both model parameters and the computational solutions built around them. Collaborative learning typically optimizes fixed models, whereas automated model design and research generally rely on centralized training and evaluation signals. Bridging the two is nontrivial: distributed evidence must support not only optimization, but also the evaluation and refinement of computational changes that generalize beyond their local settings. To tackle this, we introduce AI4AI4Cell, a collaborative research framework that couples client-local parameter fitting with an evidence-guided proposal–evaluation loop. Local training shards update a shared predictor, while local development shards produce aggregate evidence cards that guide the next executable proposal. Across three biological settings spanning drug–target interaction prediction, observed-response proteomics, and cell perturbation, the complete system leads four of five main-table endpoints. Under matched data, candidate and compute budgets, evidence-guided allocation yields four held-out wins, eight ties and no losses across twelve task–seed comparisons, including a 10.84-point mean Top-1 gain on double-gene perturbations. AI4AI4Cell offers a practical path toward self-improving biological AI under data-sharing constraints.

1. INTRODUCTION

Biological data are often distributed across institutions and data holders, where privacy and data-sharing constraints limit centralized access (Sheller et al., 2020; Bakhtiari et al., 2025). Combining these measurements could improve models of molecular interactions, protein responses and cellular perturbations. Exploiting new measurements, however, involves more than refitting parameters: researchers also choose prediction heads, adjust optimization settings and decide which candidate models deserve further training. Automating this repeated process of proposing, testing and revising computational designs motivates self-improving biological AI. The challenge is to support that process when each laboratory retains its own data.

Figure 1 contrasts two complementary approaches. A centralized research agent iterates on data accessible to its execution environment, using experiment outcomes to revise candidate solutions. AIDE and the AI Scientist automate this process for computational research (Jiang et al., 2025b; Lu et al., 2024); DrugEvolve applies algorithm revision, evaluation and memory to drug-development tasks (Zhou et al., 2026; GAIR-NLP DrugEvolve contributors, 2026). These workflows turn measurements into design feedback. Collaborative learning instead connects several data holders through local training and model-update aggregation (McMahan et al., 2017; Sheller et al., 2020; Wang et al., 2024). Here, a *client* is a laboratory’s computational worker, which operates on data held by that laboratory. This approach broadens the evidence available for fitting a specified model. Connecting the two requires a second shared interface: laboratories must also evaluate candidate designs and supply feedback that the research agent can use to choose its next experiment.

AI4AI4Cell couples these interfaces in a two-level research cycle. For any candidate design, an inner training process fits a shared predictor using laboratory-local data. An outer research loop uses the resulting development evidence to propose and select subsequent designs. In our implementation, a design specifies the trainable prediction head and its learning rate, regularization and aggregation set-

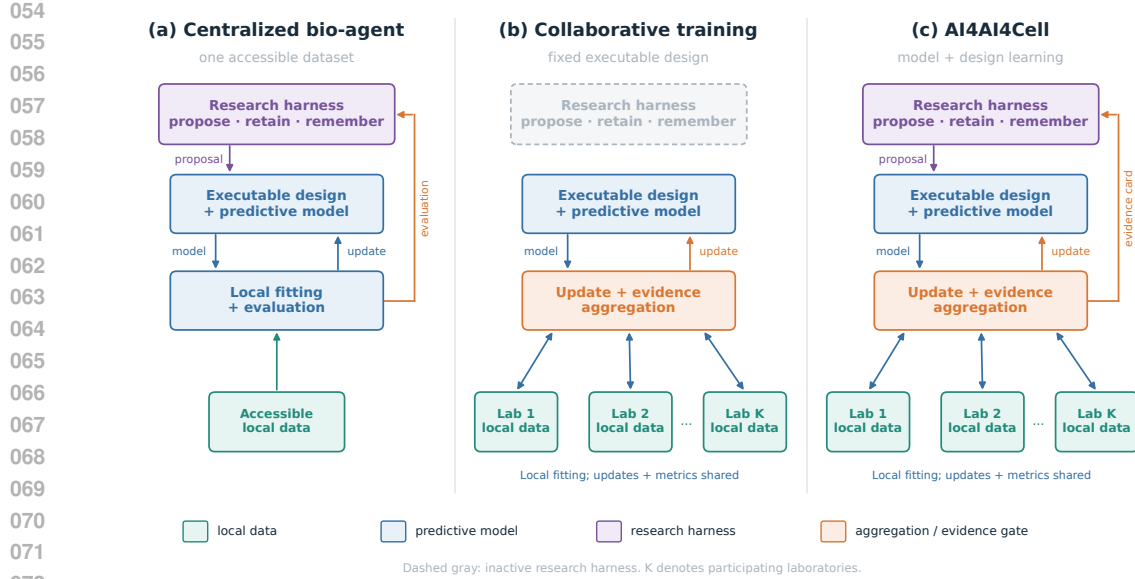


Figure 1: Three ways to use biological measurements. A centralized bio-agent revises a predictor using one accessible dataset. Collaborative training fits a specified model using data held by K laboratories. AI4AI4Cell also returns aggregate development evidence to the research loop, allowing distributed measurements to guide both parameter fitting and design revision. Colors identify data, models, research components and aggregation interfaces.

tings. The loop packages each hypothesis, executed configuration, learning trajectory and selection outcome into an *evidence card*. Accumulated cards condition the next proposal; early trajectories can also guide which candidates receive full training. Local measurements thus contribute both parameter updates and evidence for research decisions, while the language model receives aggregate diagnostics rather than raw biological responses.

We evaluate the framework in three biological settings: drug–target interaction (DTI) prediction, observed-response proteomic efficacy classification, and cell perturbation identification spanning single-gene, double-gene and drug interventions. The main comparison uses ten simulated laboratories for the collaborative system and one for its local references. Controlled comparisons then distinguish the effects of data access, design search and feedback-dependent allocation of training compute.

Our contributions are threefold. First, we connect laboratory-local parameter fitting to distributed evaluation of executable model designs through a common biological research interface. Second, we develop an evidence-guided loop mechanism that packages proposal outcomes into reusable research history and uses measured learning trajectories to select candidates for further validation. Third, we evaluate the shared mechanism across three task families, with matched-budget comparisons and sensitivity studies of laboratory participation, data sources and proposal budget. The contribution is this coupling and its experimental evaluation, with task-specific predictors connected through the same training and research interfaces.

2. RELATED WORK

Collaborative learning for biological data. Parameter aggregation enables institutions to train a shared model while retaining their measurements locally (McMahan et al., 2017; Sheller et al., 2020). In single-cell analysis, scFed addresses cell-type classification and FedscGen addresses distributed batch correction (Wang et al., 2024; Bakhtiari et al., 2025). Tabula and its Chiron implementation support collaborative single-cell foundation-model training through an agent-enabled platform (Ding et al., 2026; Mechtel et al., 2026). Automated design can interact with local training: FedEx tunes hyperparameters while training shared weights (Khodak et al., 2021), and Helmsman synthesizes

collaborative-learning systems through planning, coding and simulation-based refinement (Li et al., 2025). AI4AI4Cell connects parameter fitting to a second laboratory interface for evaluating proposed designs. Aggregate development evidence supports both proposal revision and allocation of further training.

AI for AI in biological research. Research agents turn experiment outcomes into subsequent computational proposals. AIDE searches candidate programs through drafting, debugging and improvement (Jiang et al., 2025b), while the AI Scientist connects ideas to executed experiments, replanning and writing (Lu et al., 2024). DrugEvolve applies research, implementation, evaluation and algorithm memory to drug-development tasks (Zhou et al., 2026; GAIR-NLP DrugEvolve contributors, 2026). These systems motivate executable revision and persistent experiment history. Our study asks how such feedback can be supplied by multiple data-owning laboratories through a shared research interface. We instantiate that interface with TAPB for DTI, a ProteinTalks-derived efficacy head for proteomic responses, and scDEBART-based cell representations (Lin et al., 2025; Sun et al., 2026; Sung & Kim, 2026). These models serve as task-specific prediction components; the proposal schema, local training service and evidence-driven selection mechanism remain common across tasks. The strong performance of simple references on some perturbation benchmarks further motivates separating the effect of a research loop from the choice of predictor (Ahlmann-Eltze et al., 2025).

3. METHOD: COLLABORATIVE AGENTIC BIOLOGICAL RESEARCH

3.1 FRAMEWORK OVERVIEW

AI4AI4Cell takes a task specification, an initial predictor, an editable design space and data held by K participating laboratories. The task specification defines the prediction inputs and outputs, scorer and computational budget. Each laboratory maintains separate training and development partitions. A research controller proposes a design, a shared training service fits its parameters across laboratories, and local evaluators return development measurements to a coordinator. The coordinator selects a predictor and records the evidence used in that decision. The outputs are the selected design with its fitted parameters and an ordered history of proposals and measured outcomes (Figure 2).

The key mechanism is the return of experimental evidence to the research process. An *evidence card* links a proposed change to its executed configuration, aggregate learning trajectory and selection outcome. Cards from successive experiments form a persistent history, which allows the controller to revise its next hypothesis using both successful and unsuccessful trials. The same trajectory information can guide training allocation: a candidate with promising early progress can be promoted for full validation. Thus the outer loop decides what design to investigate and how to allocate further evaluation, while the inner process fits parameters for a given design.

3.2 EVIDENCE-GUIDED PROPOSAL AND REVISION

The controller receives the task specification, current design and accumulated evidence cards. It returns a structured proposal with four fields: a `hypothesis` stating why a change should help, an `experiment` describing the proposed comparison, an `expected_effect` identifying the predicted metric or diagnostic response, and a `design_id` resolving to an executable configuration. A validator checks that the identifier is available in the task’s editable design space. The configuration determines the actual intervention; the other fields state its rationale and anticipated outcome. Appendix A gives the JSON schema and implemented design library.

After fitting, local evaluators return primary scores, losses and learning-curve diagnostics. The coordinator packages these measurements with the proposal, selected checkpoint and acceptance decision. It retains the candidate if its mean development score exceeds the incumbent’s, using lower mean loss to resolve ties. Each subsequent loop proposal receives the complete sequence of evidence cards. This separates the language model’s explanation of a candidate from the measured basis for retaining it and preserves the outcomes of rejected directions for later proposals. The research language model remains fixed; adaptation occurs in the predictive model and in the designs explored by the loop.

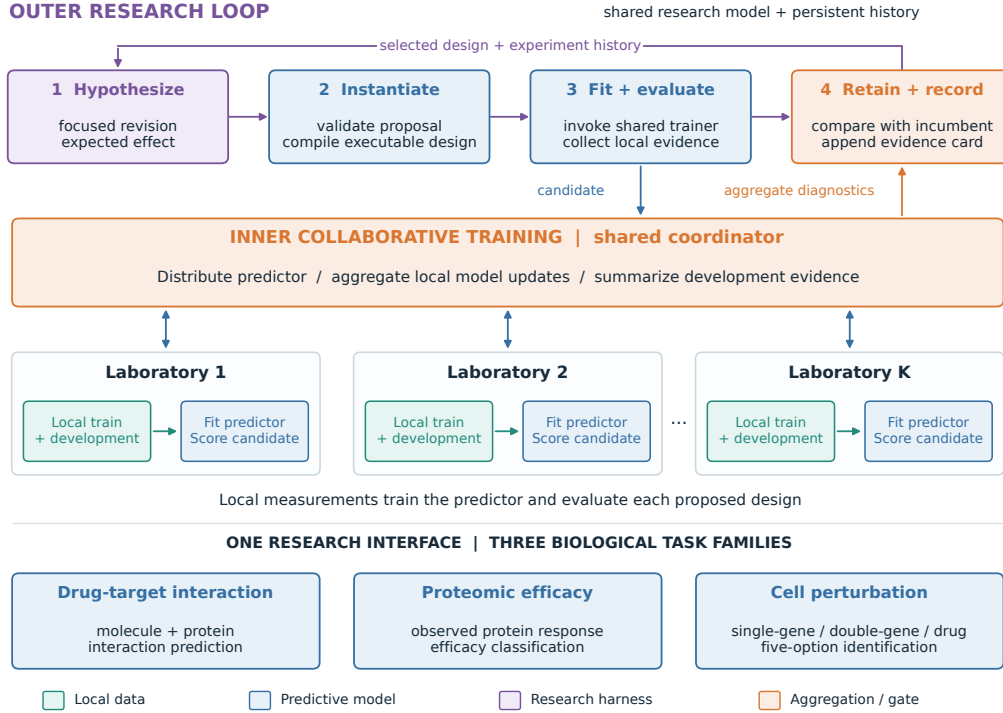


Figure 2: Shared research cycle for K laboratories. The controller converts a hypothesis into an executable design. Laboratory workers fit and evaluate it locally; the coordinator aggregates model updates and development diagnostics. The resulting evidence card supports model selection and the next research proposal. Task adapters connect the same cycle to DTI, proteomics and cell perturbation.

Evidence can influence two different research decisions. In *proposal revision*, measured history conditions which design is proposed next. In *training allocation*, short learning trajectories determine which proposed designs receive full validation. We implement the latter as a screening–promotion–validation policy that retains candidates with strong current scores as well as candidates with improving trajectories. These two decisions share the evidence interface but are evaluated separately: proposal-history comparisons assess design selection, whereas a fixed-slate racing comparison isolates evidence-dependent allocation. The experimental budgets and promotion rule are specified in Section 4 and Appendix A.

3.3 LOCAL FITTING AND DISTRIBUTED EVALUATION

For a selected design, all laboratory workers start from the same shared parameters and fit them on their local training partitions. The coordinator combines the returned parameters using training-example weights, following FedAvg (McMahan et al., 2017), with optional server momentum specified by the design. The resulting predictor is distributed for the next training round. This training pipeline is unchanged across task families; task adapters define the input representation, prediction head and loss.

Development evaluation gives each participating laboratory equal weight. If $s_i(a)$ is laboratory i ’s development score for fitted design a , the selection score is $S(a) = K^{-1} \sum_{i=1}^K s_i(a)$. Training-example weighting therefore determines how parameter updates are combined, whereas equal-laboratory weighting determines how designs and checkpoints are compared. Every candidate starts from the same task-specific initialization for its training seed. At completion, the selected checkpoint is fixed and evaluated by the task’s held-out scorer.

Workers return parameter updates and example counts for training, and scalar scores and diagnostics for research feedback. The language model receives the aggregate evidence cards. Raw interaction

rows, proteomic profiles and expression responses remain at the worker interface. The experiments simulate these roles on one host; K is an experimental parameter, with the main comparison using $K = 10$ and the sensitivity study varying K .

4. EXPERIMENTAL DESIGN

4.1 SHARED PROTOCOL AND COMPARISON METHODS

We evaluate whether collaborative data access and iterative research improve the same task-specific predictor. Training and development groups are assigned to ten disjoint simulated laboratories, balanced by compound or intervention identity. The single-laboratory setting uses preselected client 0; the collaborative setting uses all ten clients. Each comparison uses the same held-out scorer and candidate rosters.

Benchmark contracts.

Setting	Prediction contract	Task-reference model	Primary metric
DTI	Compound + protein $\rightarrow$ interaction	TAPB	AUROC $\uparrow$
Proteomics	Proteome + drug $\rightarrow$ efficacy	ProteinTalks-derived head	AP $\uparrow$
Cell (3 strata)	Response + 5 options $\rightarrow$ identity	Corrected scDEBART head	Macro Top-1 $\uparrow$

The cell contract is instantiated for VCC single-gene, Norman double-gene and Tahoe drug perturbations; task-specific held-out units and secondary metrics are specified below.

The main table compares three methods. The *task model* trains the reference architecture above with a fixed optimization recipe on one laboratory’s data. *Qwen direct* uses Qwen2.5-7B-Instruct to choose model configurations without receiving evaluation history; these configurations are trained on the same one laboratory. *AI4AI4Cell* uses the same proposer with evidence-card feedback and all ten laboratories. The language model proposes configurations; the task-specific predictor produces the biological predictions in every method. Direct and loop search share a twelve-design library, six proposal slots and the first proposal. Each candidate receives 100 training rounds, with development evaluation every five rounds. Appendix A records optimizer settings and the proposal interface.

Two controls separate the causes of improvement. A factorial comparison crosses one/ten laboratories with fixed/direct/loop research, using common seeds 42–44. It tests participation while holding research mode fixed, and research while holding participation fixed. A separate *compute-matched racing* comparison tests whether feedback improves training allocation: both methods receive the same ten designs and all ten laboratories. Uniform allocation gives each design 80 rounds. Evidence-guided allocation screens each for 20 rounds and restarts six promoted designs for 100-round validation. Both spend 800 full-client rounds and 160 development measurements. This experiment uses seeds 53–55 for proteomics and the three cell settings; its fixed candidate slate removes proposal generation as a varying factor.

Checkpoint selection uses participating laboratories’ development data, with equal weight per laboratory. Every selected predictor is evaluated on the common held-out test set. Means and sample SD summarize the specified training/search seeds on fixed partitions. Appendix A supplies all six configurations, secondary metrics and uncertainty estimates; bold marks all best displayed task scores, including ties.

4.2 DRUG–TARGET INTERACTION PREDICTION

DTI predicts a binary interaction from a compound and a protein sequence. We use the released BindingDB training/development split of 50,149/5,604 pairs. Canonical-compound groups are assigned to clients with a label-independent size-balancing rule, producing compound-disjoint clients while preserving the source split within each client.

The task reference uses the released TAPB architecture with frozen public ESM2 protein features and freshly initialized drug, fusion and classification modules (Lin et al., 2025). A target dictionary derived from client 0’s training proteins is shared identically by every arm, ensuring a common initialization. The objective combines TAPB’s classification NLL and masked-language loss under

the shared batch-64 training pipeline. Candidate designs can add a zero-output residual adapter before TAPB’s native confounder-averaged prediction.

Selection uses mean-client AUROC; AP, MCC, accuracy at 0.5 and binary cross-entropy are secondary. The fixed test suite contains 5,505 random, 3,791 unseen-drug and 2,507 unseen-protein pairs. Random-split AUROC is the main-table endpoint, and both unseen-entity endpoints appear in the complete results. All six arms and three seeds complete the full protocol.

DrugBAN supplies an additional architecture reference under the uniform multi-laboratory trainer (Bai et al., 2023; DrugBAN contributors, 2026).

4.3 PROTEOMICS-BASED EFFICACY PREDICTION

We construct an observed-response efficacy benchmark from a subset of the public ProteinTalks PTPC supplement, which provides measured proteomic responses and binary efficacy labels (Sun et al., 2026). The task predicts efficacy from a treated condition’s observed 6h/24h proteome and drug representation. We partition the 491-condition source pool, spanning 333 compounds, into 386 fitting and 105 development conditions by compound group. Ten clients each own 49–50 total conditions, with compound-disjoint training and development groups and both classes in every local development set. Local fitting sets contain 35–42 conditions.

The task reference is a ProteinTalks-derived observed-response efficacy head: protein and drug convolutional branches followed by a 64–32–1 classifier, trained from random initialization on observed 6h/24h responses and Morgan features. A 24h linear/Morgan model provides a complementary non-neural reference.

Average precision (AP) is primary; AUROC, binary cross-entropy and accuracy at 0.5 are secondary. The 148-condition test role is evaluated after the development-selected checkpoint is frozen.

4.4 CELL PERTURBATION IDENTIFICATION

Given an observed expression response and five candidate interventions, the model selects the true perturbation by cosine agreement with predicted responses. We construct three inverse-identification benchmarks: VCC-derived single-gene perturbations from the public 2025 training release with internal intervention-disjoint partitions (Arc Institute, 2025); Norman double-gene overexpression (Norman et al., 2019); and a Tahoe-100M subset of 307 compounds in A549 at 5 μ M, from the February 25, 2025 release (Zhang et al., 2025). Whole-intervention train/development/test counts are respectively 60/20/20, 70/20/27 and 180/60/67. Each laboratory owns six/seven/eighteen training interventions and two/two/six development interventions. Assignment balances intervention identities, while retaining the naturally varying number of VCC cells per intervention.

The public CELLxGENE-pretrained scDEBART backbone is frozen and its response MLP is trainable (Sung & Kim, 2026). Three upstream attention masks are implemented as additive blocking masks and verified by response-free synthetic tests. This correction leaves pretrained weights, targets, splits and candidate rosters unchanged; features are then re-extracted for every arm.

Genetic tasks use the released perturbation encoding and exclude directly perturbed genes from loss and ranking coordinates. Tahoe adds fixed Morgan-projection conditioning for chemical identities. The uniform cell objective combines response MSE with 0.1 times a contrastive loss formed from local training conditions.

Primary scoring is intervention-macro Top-1, with a 20% random-choice expectation; micro Top-1, MRR, Top-3 and cosine-logit cross-entropy are secondary. All arms receive the same five candidate interventions for each response.

4.5 PARTICIPATION, SOURCE AND SEARCH-BUDGET SENSITIVITY

We vary laboratory count over $K \in \{1, 2, 5, 10\}$ in two ways: adding the first K original clients increases available data, whereas merging all ten clients into K groups keeps the data pool fixed. Both use the default recipe and original development scoring units. A source-transfer study instead adds external collections to the complete target training pool, retaining target-only development selection and a common target test set. For VCC, the sources are the Replogle, Nadig and Jiang studies, which

provide independent perturbation collections. For proteomics, the sources are three cell contexts from the ProteinTalks mtPTDS collection; these are reported as within-study context transfer rather than as independent studies. Three fixed seeds, 61–63, are specified for both studies.

A separate proposer-budget study compares Qwen2.5-7B-Instruct with gpt-5.6-1una, using the same 36-design space and 24 proposal slots. Each slot allows two schema-validation attempts; a failed slot consumes the budget. Fixed prefixes measure the selected predictor after different amounts of search. Development trajectories, valid-proposal counts and held-out scores are reported separately in Appendix B, which also identifies completed and pending evaluations.

5. RESULTS

5.1 MAIN HELD-OUT PERFORMANCE

Table 1 places the three task families in one comparison: DTI and proteomics each contribute one endpoint, and cell perturbation contributes three. Each cell reports mean $\pm$ sample SD over seeds 42–44. This SD measures training/search variability on fixed partitions. The overall column averages the five displayed endpoints within each seed before summarizing across seeds; it is a descriptive summary, not a common biological metric.

Method	DTI	Proteomics	Cell perturbation			Overall
	TAPB AUROC $\uparrow$	PTPC AP $\uparrow$	VCC Top-1 $\uparrow$	Norman Top-1 $\uparrow$	Tahoe Top-1 $\uparrow$	Mean 5 endpoints
Task model (1 lab)	82.83 $\pm$ 1.22	25.05 $\pm$ 4.56	29.94 $\pm$ 0.00	25.56 $\pm$ 0.00	19.40 $\pm$ 0.00	36.55 $\pm$ 0.82
Qwen direct (1 lab)	<u>89.89</u> $\pm$ 1.02	20.13 $\pm$ 0.07	21.09 $\pm$ 4.38	14.15 $\pm$ 0.00	15.42 $\pm$ 6.21	32.14 $\pm$ 2.02
AI4AI4Cell (10 labs)	93.76 $\pm$ 1.11	36.99 $\pm$ 0.22	31.54 $\pm$ 0.08	<u>22.25</u> $\pm$ 0.45	22.39 $\pm$ 2.59	41.39 $\pm$ 0.60

Table 1: Main held-out comparison. Task models are TAPB (DTI), a ProteinTalks-derived efficacy head (PTPC) and corrected scDEBART response heads (cells). Qwen direct proposes configurations without evaluation history; AI4AI4Cell adds collaborative access and evidence feedback. Values are mean $\pm$ sample SD, multiplied by 100; SD is not a confidence interval. Overall summarizes the five endpoints within each seed. Bold and underline mark the best and second-best displayed means. The shared training protocol is in Section 4; all six configurations, seed counts and secondary metrics are in Appendix A.

AI4AI4Cell leads four of five displayed endpoints. TAPB reaches 93.76% random-test AUROC versus 89.89% for one-laboratory direct optimization, and proteomic AP reaches 36.99% versus 20.13%. Relative to direct optimization, VCC, Norman and Tahoe macro Top-1 increase by +10.45, +8.10 and +6.97 points. Norman’s best displayed result remains the one-laboratory fixed recipe at 25.56%, compared with 22.25% for AI4AI4Cell. The complete system’s five-endpoint mean is 41.39, versus 36.55 for the fixed recipe and 32.14 for direct optimization. The main comparison measures the combined benefit of collaborative access and research; the following controls separate them.

5.2 DOES FEEDBACK IMPROVE THE ALLOCATION OF TRAINING COMPUTE?

This experiment asks whether early development measurements help allocate a fixed training budget. The candidate designs and laboratory data are identical. Uniform allocation trains every design equally; the evidence-guided loop screens all designs and promotes a subset using current performance and learning progress. Thus Table 2 measures training allocation, independently of an LLM’s ability to generate better designs.

Across 12 newly executed proteomics/cell task–seed comparisons, the loop wins four and ties eight. Norman improves from 11.48% to 22.32% macro Top-1, a +10.84-point mean effect with hierarchical paired-bootstrap 95% CI [4.54, 17.73]. Tahoe improves by +1.99 points, with one win and two ties; VCC and proteomics tie. On Norman, a promoted low-learning-rate residual design continues improving beyond the uniform method’s stopping point. Concentrating full validation on promising candidates therefore improves the final predictor at the same total training budget. The DTI develop-

Benchmark	Uniform	Evidence-guided	Gain	W/T/L
Proteomics: efficacy	37.47	37.47	+0.00	0/3/0
Cell: VCC single-gene	20.01	20.01	+0.00	0/3/0
Cell: Norman double-gene	11.48	22.32	+10.84	3/0/0
Cell: Tahoe drug	20.90	22.89	+1.99	1/2/0

Table 2: Does early evidence improve training allocation? Held-out scores use the same ten designs, ten laboratories, 800 full-client rounds and 160 development measurements per method. Uniform allocation gives every design 80 rounds; the evidence-guided loop screens all designs and fully validates six promoted candidates. Gain is evidence-guided minus uniform, in percentage points; W/T/L counts seed-level wins, ties and losses (total 4/8/0). Norman 95% CI: [4.54, 17.73]; Tahoe: [−1.00, 7.46]. The separate DTI development replay is in Appendix A.

ment replay is reported separately in Appendix A because it reuses existing trajectories rather than newly scored held-out runs.

The factorial comparison addresses a different question: what changes when more laboratories participate? Holding the research loop fixed, ten versus one laboratory adds +3.87 AUROC points for DTI, +16.85 AP points for proteomics, and +11.32, +8.10 and +6.97 Top-1 points for the three cell settings. Appendix A reports all six combinations, including direct-versus-loop contrasts at fixed access. At six proposals, the original direct and loop configurations select the same checkpoints, so the racing result concerns evidence-dependent allocation rather than an independently demonstrated final-score gain from proposal memory.

5.3 WHAT CHANGES WITH MORE LABORATORIES, SOURCES AND PROPOSALS?

The laboratory-count study separates data availability from partitioning. With the complete pool retained, PTPC AP ranges from 34.59 at $K = 1$ to 39.19 at $K = 5$ and 37.54 at $K = 10$. Norman instead scores 49.63 at $K = 1$ and 15.48 at $K = 10$ under the default recipe. Merging clients changes both local optimization and, for cell tasks, the contrastive negative pool. These results make laboratory count a meaningful design parameter rather than a setting whose benefit can be assumed to increase monotonically.

External sources have different effects on the same target. Adding Jiang data raises VCC Top-1 from 29.49 to 33.07; adding all three external studies gives 27.33. The Jiang difference is +3.58 points with a paired intervention-bootstrap 95% CI of [−2.11, 9.51]. For proteomics, combining all three mtPTDS contexts changes AP from 34.59 to 35.47. Source compatibility, rather than source count alone, therefore deserves explicit evaluation. Appendix B retains every source combination and its uncertainty.

Longer search also distinguishes development efficiency from held-out improvement. In the completed Norman seed-61 trace, Luna reaches its best development primary score at proposal slot 4 with feedback and slot 22 without feedback, but both finish at 23.11% test Top-1 after 24 slots. Qwen likewise finishes tied at 22.30%. On VCC seed 61, both Qwen modes fall from 31.50% test Top-1 at six slots to 19.61% at 24 despite improved development performance. The ongoing multi-seed study reports proposal validity and prefix curves alongside scores, making it possible to distinguish execution reliability, faster search and improved generalization.

6. CONCLUSION

AI4AI4Cell connects laboratory-local parameter fitting to an evidence-guided research loop. The same proposal, training and evaluation interfaces support DTI, observed-response proteomics and cell perturbation. The complete system leads four of five endpoints in the main comparison, and the compute-matched racing experiment improves allocation of a fixed training budget, including a 10.84-point mean Top-1 gain on Norman. These results establish a practical way for distributed biological measurements to inform both predictive parameters and computational research decisions.

REPRODUCIBILITY STATEMENT

Section 3 specifies the research and training interfaces; Section 4 defines task inputs, data partitions, reference models and metrics. Appendix A provides the executable proposal schema, complete design library, optimization settings and matched-budget promotion rule. Appendix B specifies the laboratory-count, source-transfer and proposer-budget studies. The accompanying artifact records code and data hashes, random seeds, proposals, development trajectories, selection decisions and saved predictions. Aggregate tables are generated from completed evaluation records, with independent score recomputation for the main comparison. These records connect each reported result to its configuration and selected checkpoint.

ETHICS STATEMENT

This computational study uses previously released biological resources under their source access conditions and collects no new measurements or participant data. Laboratory boundaries are simulated by local workers. Retaining raw data at workers avoids direct transfer to the research model; parameter updates and aggregate diagnostics would still require an explicit privacy and security policy in a multi-institutional deployment. Predictions are evaluated as research outputs, not as validated biological or therapeutic interventions.

AI USE STATEMENT

Local Qwen2.5-7B-Instruct generates proposals in the main study; the proposer sensitivity study additionally uses the service identified as gpt-5.6-luna. AI assistants contributed to hypothesis refinement, experimental protocol and software development, data processing and checks, result analysis and interpretation, literature retrieval, figures, manuscript drafting and revision, and the Chinese reading companion. The research artifacts retain the generated proposals and their measured outcomes. The authors are responsible for verifying the final methods, results, citations and manuscript, including AI-assisted material.

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APPENDIX A. UNIFORM COLLABORATIVE RESEARCH PROTOCOL

Versioned implementation. The reported study uses `core_v3.py`, `run_v3.py` and `research_v2.py`. All task families execute the same optimization, aggregation, proposal and retention source through task-specific adapters. TAPB workers use the TAPB environment; Qwen, proteomics and cell adapters use the LDA environment. Each scientific fit starts from its task-specific public or random initialization, with no reuse of earlier task-trained weights.

Shared optimization and checkpoint selection. Each round uses one local AdamW epoch, batch size 64 and gradient-norm clipping at one; local optimizer state resets between rounds. Parameters are averaged with training-example weights. If server momentum is enabled, the coordinator multiplies the previous velocity by the configured momentum coefficient and adds the difference between the averaged parameters and the previous shared parameters, then adds that velocity to the shared parameters. Floating model buffers are averaged and integer buffers use an elementwise maximum. Momentum acts on trainable parameters only. Every full-training candidate runs for 100 rounds; local development scores are collected every five rounds. Checkpoint selection maximizes the equal-client primary score, then minimizes mean development loss. The optional residual adapter has width 32 and zero-initialized output. The DTI adapter trains TAPB using frozen public protein features, the proteomic adapter trains its efficacy head from random initialization, and the cell adapters train response heads on frozen corrected scDEBART features.

Complete design library. Each identifier below compiles to one executable candidate. D00 is the fixed starting recipe; D01–D11 are available to both direct and loop search without within-arm repetition. Six proposal slots define the common budget.

Design	Learning rate	Weight decay	Momentum	Residual adapter
D00	0.001	0.001	0	no
D01	0.0001	0.001	0	no
D02	0.01	0.001	0	no
D03	0.001	0	0	no
D04	0.001	0.01	0	no
D05	0.001	0.001	0.9	no
D06	0.001	0.001	0	yes
D07	0.001	0.001	0.9	yes
D08	0.0001	0.001	0	yes
D09	0.0001	0.001	0.9	yes
D10	0.01	0.01	0	no
D11	0.001	0.01	0	yes

Proposal generation and evidence memory. The main-study proposer is Qwen2.5-7B-Instruct, using bfloat16 inference, temperature 0.7, top- p 0.9 and at most 512 new tokens. Each slot uses a deterministic seed derived from the training seed. Direct and loop comparisons share the same first proposal. Later direct proposals receive the default reference configuration and remaining untried identifiers, without evaluation scores. Loop proposals instead receive the latest retained configuration and ordered evidence cards, each containing the candidate configuration, aggregate best development metric and loss, best round, first/last training and validation records, and final status. A bounded second generation attempt repairs schema or novelty violations. Failed proposals consume a slot. Identical configurations within a seed and participation setting can reuse a completed fit while retaining their logical positions in the research traces.

Every proposal has the following executable JSON shape:

```
{
  "hypothesis": "one testable claim about an editable factor",
  "experiment": "the controlled change and incumbent comparison",
  "expected_effect": "the expected metric and diagnostic response",
  "design_id": "D00--D11"
}
```

The compiler validates `design_id` against the table above. The finalizer then attaches executed measurements and a retention decision to the proposal, creating the evidence card used in the next iteration.

Executed loop trajectories. Three ten-laboratory traces illustrate how the common finalizer operates. Values are aggregate development measurements on the 0–1 primary-metric scale, and proposal slots are one-indexed.

Study	Selected design	Primary	Loss	First direct / loop slot
PTPC, seed 42	D04	0.665833	0.521161	5 / 5
VCC, seed 43	D00	0.240350	1.607792	initial baseline
Norman, seed 42	D08	0.519000	1.538505	3 / 2

PTPC selects stronger weight decay through the BCE tie-break at equal AP. VCC retains the initial design after six alternatives. On Norman, the loop reaches the winning lower-learning-rate residual design one slot before direct search. Across all 15 ten-laboratory task–seed trajectories, the two-proposal loop is better in two, tied in thirteen and worse in none; the mean development-primary difference is +0.78 percentage points. At six proposals, direct and loop select the same checkpoints, reflecting coverage saturation in the twelve-design library.

Compute-matched evidence racing. The follow-up feedback-isolation protocol uses a ten-design fractional-factorial slate and newly executed seeds 53–55. Direct search assigns 80 rounds to every design. The loop assigns 20 screening rounds to every design, stratifies candidates by learning rate, and promotes both the strongest current candidate and the fastest-improving candidate within each stratum. Six promoted candidates restart from the common initialization for 100-round validation. Consequently, each arm executes 800 full-client rounds and 160 development measurements over the same ten hypotheses. The promotion record is finalized before any held-out response is read.

Across the resulting 12 proteomics/cell held-out comparisons, racing gives four wins, eight ties and no losses. Norman improves by 10.84 Top-1 points on average, with seed-level effects of +10.00, +11.70 and +10.81 and a hierarchical paired-bootstrap 95% interval of [4.54, 17.73]. Tahoe gives one +5.97-point seed and two ties; VCC and PTPC tie in all three seeds. Independent rescoring of every saved prediction reproduces the reported metrics exactly. A TAPB companion replay applies the same screen/promote principle to the seven-design intersection of existing traces: direct receives 525 candidate-rounds, the loop receives 520, and the development AUROC effect is +0.39 points on average (two wins, one tie).

Client and split integrity. DTI clients contain 4,932–5,058 training and 517–643 development pairs. Compound identities do not cross clients. The fixed test roles contain 5,505 random, 3,791 unseen-drug and 2,507 unseen-protein pairs. PTPC partitions 333 compound groups into 386 fitting and 105 development conditions, with 35–42 fitting conditions per client; its frozen test role contains 148 conditions. VCC, Norman and Tahoe use whole-intervention splits of 60/20/20, 70/20/27 and 180/60/67. Their ten clients own equal numbers of training and development intervention identities.

Proteomic preprocessing. For each 6h and 24h profile, the PTPC adapter applies log1p, within-profile min–max normalization over finite observations, zero filling and unit L2 normalization. The 2,048-bit Morgan vector is separately L2-normalized and scaled by 0.5. Concatenating the two 5,171-dimensional proteomic blocks with the drug vector produces a 12,390-dimensional float32 input. The ProteinTalks-derived efficacy head contains 464,193 parameters at D00. Every arm uses identical inputs, initialization policy, 100-round budget and checkpoint rule.

Full factorial results. The following tables report all six held-out arms, complete method contrasts, proposal-prefix analysis, secondary metrics, uncertainty and all three DTI test roles. Bold marks the best displayed value within each task and metric, including ties. Higher is better except BCE and cross-entropy.

Labs	Method	DTI Proteomics		Cell perturbation		
		AUROC ↑	AP ↑	VCC Top-1 ↑	Norman Top-1 ↑	Tahoe Top-1 ↑
1	Fixed model	82.83±1.22	25.05±4.56	29.94±0.00	25.56 ±0.00	19.40±0.00
1	Direct	89.89±1.02	20.13±0.07	21.09±4.38	14.15±0.00	15.42±6.21
1	Our loop	89.89±1.02	20.13±0.07	20.22±2.22	14.15±0.00	15.42±6.21
10	Fixed model	91.80±2.55	37.07 ±0.08	31.54 ±0.08	15.48±0.00	25.87 ±1.72
10	Direct	93.76 ±1.11	36.99±0.22	31.54 ±0.08	22.25±0.45	22.39±2.59
10	Our loop	93.76 ±1.11	36.99±0.22	31.54 ±0.08	22.25±0.45	22.39±2.59
Completed seeds ($n/3$)		3/3	3/3	3/3	3/3	3/3

Table A.1: Six-arm ablation of the task-reference models. Entries are held-out mean $\pm$ sample SD across completed training/search seeds, multiplied by 100. The data partitions remain fixed; this SD measures run-to-run variation, not dataset uncertainty. DTI uses the random held-out endpoint. Bold marks all displayed maxima.

Held-out contrast	DTI	Proteomics	VCC	Norman	Tahoe
Loop – fixed (1 lab)	+7.06	-4.91	-9.71	-11.41	-3.98
Loop – fixed (10 labs)	+1.96	-0.08	+0.00	+6.77	-3.48
Loop – direct (1 lab)	+0.00	+0.00	-0.87	+0.00	+0.00
Loop – direct (10 labs)	+0.00	+0.00	+0.00	+0.00	+0.00
10 – 1 labs (both fixed)	+8.97	+12.02	+1.61	-10.07	+6.47
10 – 1 labs (both direct)	+3.87	+16.85	+10.45	+8.10	+6.97
10 – 1 labs (both loop)	+3.87	+16.85	+11.32	+8.10	+6.97
Completed seeds ($n/3$)	3/3	3/3	3/3	3/3	3/3

Table A.2: Complete decomposition into design-search improvement (loop minus fixed), feedback-specific improvement (loop minus matched-budget direct), and laboratory-participation improvement (ten minus one lab with loop fixed). Mean paired held-out differences are in percentage points over the three seeds and are computed before rounding.

Proposal budget	Loop better	Tie	Direct better	Mean Δ primary points
task-seed trajectories				
1	0	15	0	+0.00
2	2	13	0	+0.78
3	0	14	1	-0.04
4	0	15	0	+0.00
5	0	14	1	-0.02
6	0	15	0	+0.00

Table A.3: Executed-prefix analysis of all 15 ten-laboratory development trajectories at common proposal budgets. At two proposals, feedback-guided search improves TAPB seed 44 by 4.84 points and Norman seed 42 by 6.90 points, with thirteen ties and no regressions.

Table A.4: Held-out secondary metrics for the task-reference models, averaged over seeds 42–44. Bold indicates all tied maxima, or minima for loss.

Method	Labs	Metric 1	Metric 2	Metric 3	Metric 4
DTI ($n = 3$)		AUROC	AP	MCC	BCE
Fixed model	1	0.8283	0.8157	0.4995	0.5237
Direct	1	0.8989	0.8777	0.6571	0.6102
Our loop	1	0.8989	0.8777	0.6571	0.6102
Fixed model	10	0.9180	0.8936	0.7239	0.3654
Direct	10	0.9376	0.9273	0.7371	0.3659
Our loop	10	0.9376	0.9273	0.7371	0.3659
Proteomics ($n = 3$)		AP	AUROC	BCE	Acc.
Fixed model	1	0.2505	0.6549	0.4265	0.8514
Direct	1	0.2013	0.6180	0.4434	0.8176
Our loop	1	0.2013	0.6180	0.4434	0.8176
Fixed model	10	0.3707	0.7406	0.3978	0.8514
Direct	10	0.3699	0.7387	0.3983	0.8514
Our loop	10	0.3699	0.7387	0.3983	0.8514
VCC ($n = 3$)		Micro Top-1	MRR	Top-3	CE
Fixed model	1	0.3361	0.5653	0.7389	1.5451
Direct	1	0.2148	0.4710	0.6278	1.6082
Our loop	1	0.2074	0.4694	0.6426	1.6086
Fixed model	10	0.3398	0.5634	0.6944	1.5814
Direct	10	0.3398	0.5634	0.6944	1.5814
Our loop	10	0.3398	0.5634	0.6944	1.5814
Norman ($n = 3$)		Micro Top-1	MRR	Top-3	CE
Fixed model	1	0.2556	0.4850	0.6393	1.5874
Direct	1	0.1415	0.4212	0.5835	1.6039
Our loop	1	0.1415	0.4212	0.5835	1.6039
Fixed model	10	0.1548	0.4526	0.6563	1.5806
Direct	10	0.2225	0.5007	0.7096	1.5883
Our loop	10	0.2225	0.5007	0.7096	1.5883
Tahoe ($n = 3$)		Micro Top-1	MRR	Top-3	CE
Fixed model	1	0.1940	0.4500	0.5821	1.6308
Direct	1	0.1542	0.4260	0.5672	1.6190
Our loop	1	0.1542	0.4260	0.5672	1.6190
Fixed model	10	0.2587	0.5020	0.6517	1.6061
Direct	10	0.2239	0.4799	0.6169	1.6075
Our loop	10	0.2239	0.4795	0.6119	1.6075

Table A.5: Paired cluster-bootstrap 95% intervals for each seed and main contrast, in percentage points. Intervals condition on the selected trained models; DTI random-endpoint rows resample drug clusters.

Task	Contrast	Seed	Difference	95% interval
DTI	Loop – fixed (10 labs)	42	+0.33	[-0.06, +0.69]
DTI	Loop – direct (1 lab)	42	+0.00	[+0.00, +0.00]
DTI	Loop – direct (10 labs)	42	+0.00	[+0.00, +0.00]
DTI	10 – 1 labs (both fixed)	42	+12.51	[+11.35, +13.71]
DTI	10 – 1 labs (both direct)	42	+5.78	[+5.02, +6.57]
DTI	10 – 1 labs (both loop)	42	+5.78	[+5.02, +6.57]
DTI	Loop – fixed (10 labs)	43	+0.54	[-0.08, +1.11]
DTI	Loop – direct (1 lab)	43	+0.00	[+0.00, +0.00]

Task	Contrast	Seed	Difference	95% interval
DTI	Loop – direct (10 labs)	43	+0.00	[+0.00, +0.00]
DTI	10 – 1 labs (both fixed)	43	+7.80	[+6.82, +8.82]
DTI	10 – 1 labs (both direct)	43	+1.65	[+0.90, +2.38]
DTI	10 – 1 labs (both loop)	43	+1.65	[+0.90, +2.38]
DTI	Loop – fixed (10 labs)	44	+5.01	[+4.25, +5.71]
DTI	Loop – direct (1 lab)	44	+0.00	[+0.00, +0.00]
DTI	Loop – direct (10 labs)	44	+0.00	[+0.00, +0.00]
DTI	10 – 1 labs (both fixed)	44	+6.60	[+5.61, +7.74]
DTI	10 – 1 labs (both direct)	44	+4.17	[+3.45, +4.93]
DTI	10 – 1 labs (both loop)	44	+4.17	[+3.45, +4.93]
Proteomics	Loop – fixed (10 labs)	42	+0.00	[+0.00, +0.00]
Proteomics	Loop – direct (1 lab)	42	+0.00	[+0.00, +0.00]
Proteomics	Loop – direct (10 labs)	42	+0.00	[+0.00, +0.00]
Proteomics	10 – 1 labs (both fixed)	42	+13.16	[-0.39, +26.02]
Proteomics	10 – 1 labs (both direct)	42	+16.91	[+3.37, +31.02]
Proteomics	10 – 1 labs (both loop)	42	+16.91	[+3.37, +31.02]
Proteomics	Loop – fixed (10 labs)	43	+0.00	[+0.00, +0.00]
Proteomics	Loop – direct (1 lab)	43	+0.00	[+0.00, +0.00]
Proteomics	Loop – direct (10 labs)	43	+0.00	[+0.00, +0.00]
Proteomics	10 – 1 labs (both fixed)	43	+15.98	[+1.29, +29.00]
Proteomics	10 – 1 labs (both direct)	43	+17.03	[+3.34, +31.44]
Proteomics	10 – 1 labs (both loop)	43	+17.03	[+3.34, +31.44]
Proteomics	Loop – fixed (10 labs)	44	-0.24	[-3.40, +2.33]
Proteomics	Loop – direct (1 lab)	44	+0.00	[+0.00, +0.00]
Proteomics	Loop – direct (10 labs)	44	+0.00	[+0.00, +0.00]
Proteomics	10 – 1 labs (both fixed)	44	+6.93	[-3.43, +15.60]
Proteomics	10 – 1 labs (both direct)	44	+16.63	[+3.02, +31.08]
Proteomics	10 – 1 labs (both loop)	44	+16.63	[+3.02, +31.08]
VCC	Loop – fixed (10 labs)	42	+0.00	[+0.00, +0.00]
VCC	Loop – direct (1 lab)	42	+0.00	[+0.00, +0.00]
VCC	Loop – direct (10 labs)	42	+0.00	[+0.00, +0.00]
VCC	10 – 1 labs (both fixed)	42	+1.70	[-5.36, +8.58]
VCC	10 – 1 labs (both direct)	42	+9.90	[-2.88, +23.04]
VCC	10 – 1 labs (both loop)	42	+9.90	[-2.88, +23.04]
VCC	Loop – fixed (10 labs)	43	+0.00	[+0.00, +0.00]
VCC	Loop – direct (1 lab)	43	+4.83	[-5.50, +16.49]
VCC	Loop – direct (10 labs)	43	+0.00	[+0.00, +0.00]
VCC	10 – 1 labs (both fixed)	43	+1.56	[-5.76, +8.58]
VCC	10 – 1 labs (both direct)	43	+15.08	[+2.94, +27.78]
VCC	10 – 1 labs (both loop)	43	+10.24	[-2.74, +23.49]
VCC	Loop – fixed (10 labs)	44	+0.00	[+0.00, +0.00]
VCC	Loop – direct (1 lab)	44	-7.44	[-17.23, +1.33]
VCC	Loop – direct (10 labs)	44	+0.00	[+0.00, +0.00]
VCC	10 – 1 labs (both fixed)	44	+1.56	[-5.76, +8.58]
VCC	10 – 1 labs (both direct)	44	+6.38	[-4.38, +19.13]
VCC	10 – 1 labs (both loop)	44	+13.82	[+2.54, +26.34]
Norman	Loop – fixed (10 labs)	42	+7.26	[-9.41, +25.41]
Norman	Loop – direct (1 lab)	42	+0.00	[+0.00, +0.00]
Norman	Loop – direct (10 labs)	42	+0.00	[+0.00, +0.00]
Norman	10 – 1 labs (both fixed)	42	-10.07	[-21.19, +0.00]
Norman	10 – 1 labs (both direct)	42	+8.59	[-8.30, +26.00]
Norman	10 – 1 labs (both loop)	42	+8.59	[-8.30, +26.00]
Norman	Loop – fixed (10 labs)	43	+6.37	[-11.12, +24.81]

Task	Contrast	Seed	Difference	95% interval
Norman	Loop – direct (1 lab)	43	+0.00	[+0.00, +0.00]
Norman	Loop – direct (10 labs)	43	+0.00	[+0.00, +0.00]
Norman	10 – 1 labs (both fixed)	43	-10.07	[-21.19, +0.00]
Norman	10 – 1 labs (both direct)	43	+7.70	[-9.33, +25.85]
Norman	10 – 1 labs (both loop)	43	+7.70	[-9.33, +25.85]
Norman	Loop – fixed (10 labs)	44	+6.67	[-10.37, +24.81]
Norman	Loop – direct (1 lab)	44	+0.00	[+0.00, +0.00]
Norman	Loop – direct (10 labs)	44	+0.00	[+0.00, +0.00]
Norman	10 – 1 labs (both fixed)	44	-10.07	[-21.19, +0.00]
Norman	10 – 1 labs (both direct)	44	+8.00	[-8.52, +25.93]
Norman	10 – 1 labs (both loop)	44	+8.00	[-8.52, +25.93]
Tahoe	Loop – fixed (10 labs)	42	+0.00	[+0.00, +0.00]
Tahoe	Loop – direct (1 lab)	42	+0.00	[+0.00, +0.00]
Tahoe	Loop – direct (10 labs)	42	+0.00	[+0.00, +0.00]
Tahoe	10 – 1 labs (both fixed)	42	+4.48	[-5.97, +14.93]
Tahoe	10 – 1 labs (both direct)	42	+10.45	[-1.49, +22.39]
Tahoe	10 – 1 labs (both loop)	42	+10.45	[-1.49, +22.39]
Tahoe	Loop – fixed (10 labs)	43	-2.99	[-7.46, +0.00]
Tahoe	Loop – direct (1 lab)	43	+0.00	[+0.00, +0.00]
Tahoe	Loop – direct (10 labs)	43	+0.00	[+0.00, +0.00]
Tahoe	10 – 1 labs (both fixed)	43	+7.46	[-4.48, +17.95]
Tahoe	10 – 1 labs (both direct)	43	+13.43	[+1.49, +23.88]
Tahoe	10 – 1 labs (both loop)	43	+13.43	[+1.49, +23.88]
Tahoe	Loop – fixed (10 labs)	44	-7.46	[-19.40, +4.48]
Tahoe	Loop – direct (1 lab)	44	+0.00	[+0.00, +0.00]
Tahoe	Loop – direct (10 labs)	44	+0.00	[+0.00, +0.00]
Tahoe	10 – 1 labs (both fixed)	44	+7.46	[-4.48, +17.95]
Tahoe	10 – 1 labs (both direct)	44	-2.99	[-17.91, +11.94]
Tahoe	10 – 1 labs (both loop)	44	-2.99	[-17.91, +11.94]

Table A.6: All three frozen TAPB held-out endpoints under the committed common protocol. Scores are completed-seed means multiplied by 100; N/A means no complete independently verified six-arm evaluation. The main DTI endpoint is random held-out.

Endpoint	Method	Labs	AUROC	AP
Random	Fixed model	1	82.83	81.57
Random	Direct	1	89.89	87.77
Random	Our loop	1	89.89	87.77
Random	Fixed model	10	91.80	89.36
Random	Direct	10	93.76	92.73
Random	Our loop	10	93.76	92.73
Unseen drug	Fixed model	1	81.96	85.78
Unseen drug	Direct	1	88.39	89.35
Unseen drug	Our loop	1	88.39	89.35
Unseen drug	Fixed model	10	90.06	90.40
Unseen drug	Direct	10	91.65	92.58
Unseen drug	Our loop	10	91.65	92.58
Unseen protein	Fixed model	1	76.40	78.75
Unseen protein	Direct	1	83.26	80.53
Unseen protein	Our loop	1	83.26	80.53
Unseen protein	Fixed model	10	85.80	78.92

Endpoint	Method	Labs	AUROC	AP
Unseen protein	Direct	10	87.63	85.07
Unseen protein	Our loop	10	87.63	85.07

Table A.7: Post-hoc TAPB paired AUROC/AP differences and 95% intervals from 1,000 fixed-seed cluster-bootstrap draws. Random and unseen-drug endpoints use drug clusters; unseen-protein uses protein clusters. All three endpoints, six arms and prespecified contrasts are shown.

Endpoint / seed	Contrast	AUROC Δ	95% interval	AP Δ	95% interval
Random / 42	Loop — fixed (10 labs)	+0.33	[-0.06, +0.69]	+1.40	[+0.76, +2.05]
Random / 42	Loop — direct (1 lab)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Random / 42	Loop — direct (10 labs)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Random / 42	10 — 1 labs (fixed)	+12.51	[+11.35, +13.71]	+10.56	[+9.10, +12.11]
Random / 42	10 — 1 labs (direct)	+5.78	[+5.02, +6.57]	+8.29	[+7.01, +9.56]
Random / 42	10 — 1 labs (loop)	+5.78	[+5.02, +6.57]	+8.29	[+7.01, +9.56]
Random / 42	Our harness — fixed (1 lab)	+12.84	[+11.82, +14.01]	+11.96	[+10.56, +13.45]
Random / 42	Our harness — direct (1 lab)	+5.78	[+5.02, +6.57]	+8.29	[+7.01, +9.56]
Unseen drug / 42	Loop — fixed (10 labs)	+0.10	[-0.47, +0.67]	+1.06	[+0.42, +1.66]
Unseen drug / 42	Loop — direct (1 lab)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Unseen drug / 42	Loop — direct (10 labs)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Unseen drug / 42	10 — 1 labs (fixed)	+12.30	[+10.77, +13.98]	+7.17	[+5.82, +8.72]
Unseen drug / 42	10 — 1 labs (direct)	+6.38	[+5.34, +7.53]	+7.55	[+6.30, +8.92]
Unseen drug / 42	10 — 1 labs (loop)	+6.38	[+5.34, +7.53]	+7.55	[+6.30, +8.92]
Unseen drug / 42	Our harness — fixed (1 lab)	+12.40	[+10.94, +13.85]	+8.22	[+6.97, +9.66]
Unseen drug / 42	Our harness — direct (1 lab)	+6.38	[+5.34, +7.53]	+7.55	[+6.30, +8.92]
Unseen protein / 42	Loop — fixed (10 labs)	+0.03	[-5.53, +4.28]	+4.12	[-4.10, +8.22]
Unseen protein / 42	Loop — direct (1 lab)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Unseen protein / 42	Loop — direct (10 labs)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Unseen protein / 42	10 — 1 labs (fixed)	+20.02	[+0.16, +63.83]	+8.14	[-8.27, +51.77]
Unseen protein / 42	10 — 1 labs (direct)	+8.86	[+2.48, +13.39]	+13.26	[+3.15, +28.50]
Unseen protein / 42	10 — 1 labs (loop)	+8.86	[+2.48, +13.39]	+13.26	[+3.15, +28.50]
Unseen protein / 42	Our harness — fixed (1 lab)	+20.05	[+1.84, +61.27]	+12.25	[-3.01, +53.19]
Unseen protein / 42	Our harness — direct (1 lab)	+8.86	[+2.48, +13.39]	+13.26	[+3.15, +28.50]
Random / 43	Loop — fixed (10 labs)	+0.54	[-0.08, +1.11]	+0.44	[-0.54, +1.44]
Random / 43	Loop — direct (1 lab)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Random / 43	Loop — direct (10 labs)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Random / 43	10 — 1 labs (fixed)	+7.80	[+6.82, +8.82]	+7.68	[+6.32, +9.13]
Random / 43	10 — 1 labs (direct)	+1.65	[+0.90, +2.38]	+1.17	[-0.11, +2.50]
Random / 43	10 — 1 labs (loop)	+1.65	[+0.90, +2.38]	+1.17	[-0.11, +2.50]
Random / 43	Our harness — fixed (1 lab)	+8.35	[+7.29, +9.37]	+8.12	[+6.53, +9.76]
Random / 43	Our harness — direct (1 lab)	+1.65	[+0.90, +2.38]	+1.17	[-0.11, +2.50]
Unseen drug / 43	Loop — fixed (10 labs)	-0.51	[-1.44, +0.29]	-1.10	[-2.16, -0.18]
Unseen drug / 43	Loop — direct (1 lab)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Unseen drug / 43	Loop — direct (10 labs)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Unseen drug / 43	10 — 1 labs (fixed)	+6.68	[+5.33, +8.25]	+4.66	[+3.33, +6.36]
Unseen drug / 43	10 — 1 labs (direct)	-0.15	[-0.96, +0.67]	-1.43	[-2.43, -0.42]
Unseen drug / 43	10 — 1 labs (loop)	-0.15	[-0.96, +0.67]	-1.43	[-2.43, -0.42]
Unseen drug / 43	Our harness — fixed (1 lab)	+6.17	[+4.85, +7.48]	+3.57	[+2.10, +5.09]
Unseen drug / 43	Our harness — direct (1 lab)	-0.15	[-0.96, +0.67]	-1.43	[-2.43, -0.42]
Unseen protein / 43	Loop — fixed (10 labs)	-0.96	[-7.24, +3.07]	-1.83	[-21.55, +7.54]
Unseen protein / 43	Loop — direct (1 lab)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Unseen protein / 43	Loop — direct (10 labs)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Unseen protein / 43	10 — 1 labs (fixed)	+4.79	[-6.61, +28.12]	+0.41	[-16.48, +31.69]
Unseen protein / 43	10 — 1 labs (direct)	-1.19	[-6.22, +6.09]	-6.76	[-13.95, +6.71]
Unseen protein / 43	10 — 1 labs (loop)	-1.19	[-6.22, +6.09]	-6.76	[-13.95, +6.71]
Unseen protein / 43	Our harness — fixed (1 lab)	+3.83	[-5.83, +22.32]	-1.42	[-14.49, +16.88]
Unseen protein / 43	Our harness — direct (1 lab)	-1.19	[-6.22, +6.09]	-6.76	[-13.95, +6.71]
Random / 44	Loop — fixed (10 labs)	+5.01	[+4.25, +5.71]	+8.28	[+7.07, +9.56]
Random / 44	Loop — direct (1 lab)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Random / 44	Loop — direct (10 labs)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Random / 44	10 — 1 labs (fixed)	+6.60	[+5.61, +7.74]	+5.12	[+3.71, +6.80]
Random / 44	10 — 1 labs (direct)	+4.17	[+3.45, +4.93]	+5.44	[+4.23, +6.64]
Random / 44	10 — 1 labs (loop)	+4.17	[+3.45, +4.93]	+5.44	[+4.23, +6.64]
Random / 44	Our harness — fixed (1 lab)	+11.60	[+10.60, +12.65]	+13.40	[+11.88, +15.08]
Random / 44	Our harness — direct (1 lab)	+4.17	[+3.45, +4.93]	+5.44	[+4.23, +6.64]
Unseen drug / 44	Loop — fixed (10 labs)	+5.18	[+4.09, +6.19]	+6.57	[+5.28, +7.85]
Unseen drug / 44	Loop — direct (1 lab)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Unseen drug / 44	Loop — direct (10 labs)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Unseen drug / 44	10 — 1 labs (fixed)	+5.32	[+3.81, +6.97]	+2.04	[+0.38, +3.90]
Unseen drug / 44	10 — 1 labs (direct)	+3.54	[+2.69, +4.48]	+3.57	[+2.55, +4.72]
Unseen drug / 44	10 — 1 labs (loop)	+3.54	[+2.69, +4.48]	+3.57	[+2.55, +4.72]

Endpoint / seed	Contrast	AUROC Δ	95% interval	AP Δ	95% interval
Unseen drug / 44	Our harness — fixed (1 lab)	+10.50	[+9.20, +11.87]	+8.61	[+7.10, +10.29]
Unseen drug / 44	Our harness — direct (1 lab)	+3.54	[+2.69, +4.48]	+3.57	[+2.55, +4.72]
Unseen protein / 44	Loop — fixed (10 labs)	+6.42	[-2.79, +17.37]	+16.15	[-5.79, +26.26]
Unseen protein / 44	Loop — direct (1 lab)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Unseen protein / 44	Loop — direct (10 labs)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Unseen protein / 44	10 — 1 labs (fixed)	+3.39	[-16.02, +33.81]	-8.03	[-26.82, +16.99]
Unseen protein / 44	10 — 1 labs (direct)	+5.45	[+0.61, +8.74]	+7.10	[-8.65, +12.63]
Unseen protein / 44	10 — 1 labs (loop)	+5.45	[+0.61, +8.74]	+7.10	[-8.65, +12.63]
Unseen protein / 44	Our harness — fixed (1 lab)	+9.81	[-2.07, +34.89]	+8.12	[-12.06, +22.71]
Unseen protein / 44	Our harness — direct (1 lab)	+5.45	[+0.61, +8.74]	+7.10	[-8.65, +12.63]

Uncertainty analysis. The DTI analysis is computed after training and held-out scoring, using paired cluster bootstrap with 1,000 fixed-seed resamples of retained predictions. It groups by released drug identity on random/unseen-drug endpoints and by protein sequence on unseen-protein. Other task intervals use their natural compound or intervention grouping. Resamples are shared within each method contrast. These intervals characterize sample-level variation within each selected fit; between-training-seed variation is reported through the three-seed means and sample SD.

Verification. CPU unit and integration tests cover the common trainer, task adapters, proposal generation, frozen evaluation and cell attention-mask correction. Publication checks bind table cells to completed arms, model versions, checkpoint seals and independent score recomputation. The snapshot includes receipt hashes, aggregate metrics and the complete proposal histories used in this manuscript.

APPENDIX B. LABORATORY, SOURCE AND PROPOSER SENSITIVITY

Study status and common settings. These follow-up studies specify seeds 61–63 and retain the task predictors, scorer and inner training procedure of Appendix A. They address sensitivity rather than replacing the main-table protocol. Tables identify the number of completed evaluations at this revision; an unfinished study is N/A. The active DTI queue runs independently of the lighter head-training studies. Independent verification recomputes scores from saved predictions and checks their selected-checkpoint seals.

Number of laboratories. The participation experiment uses a fixed nested set of the first K original clients, increasing both training and development access. The fixed-pool experiment merges the complete ten-client pool into K training groups while retaining equal-weight evaluation over the original ten development panels. Thus its selection population stays fixed as training partitions change. $K \in \{1, 2, 5, 10\}$; the shared $K = 10$ fit is reused by both comparisons. Every unique fit uses the default design and 100 rounds. The cell objective forms contrastive negatives within each local training group, so repartitioning changes the negative pool as well as optimization steps. The curves characterize this implemented training procedure; they do not isolate communication alone. Zero between-seed SD can occur when a fixed pretrained head receives an identical single-batch update and does not imply zero biological uncertainty.

Task	Comparison	n	$K = 1$	$K = 2$	$K = 5$	$K = 10$
DTI	More data	0/3	N/A	N/A	N/A	N/A
DTI	Fixed pool	0/3	N/A	N/A	N/A	N/A
PTPC	More data	3/3	26.03 ± 5.57	37.38 ± 0.60	29.63 ± 2.50	37.54 ± 0.27
PTPC	Fixed pool	3/3	34.59 ± 5.66	37.36 ± 1.89	39.19 ± 0.43	37.54 ± 0.27
VCC	More data	3/3	30.02 ± 0.78	39.56 ± 0.11	31.88 ± 0.00	31.48 ± 0.16
VCC	Fixed pool	3/3	29.49 ± 0.00	28.80 ± 0.20	31.06 ± 0.22	31.48 ± 0.16
Norman	More data	3/3	25.56 ± 0.00	18.22 ± 0.00	19.68 ± 1.11	15.48 ± 0.00
Norman	Fixed pool	3/3	49.63 ± 4.20	19.56 ± 0.20	26.12 ± 0.11	15.48 ± 0.00
Tahoe	More data	3/3	19.40 ± 0.00	25.37 ± 0.00	20.90 ± 0.00	26.37 ± 0.86
Tahoe	Fixed pool	3/3	22.39 ± 0.00	21.39 ± 0.86	20.40 ± 0.86	26.37 ± 0.86

Table B.1: Laboratory-count sensitivity with the fixed default recipe. More data adds a nested set of original clients; fixed pool retains all data and merges clients. Scores are mean $\pm$ sample SD times 100. n is completed versus specified seeds. N/A indicates an incomplete evaluation, not a zero.

Independent sources and target evaluation. For transfer, the entire target training pool is one study-level client. Each auxiliary collection contributes a separate local update, using the same sample-weighted aggregation. Only the original target development panels select the checkpoint; all methods share the target test set. Each target-only, individual-source and all-source configuration is trained from the same task initialization. VCC auxiliary data comprise Replogle K562, Nadig HepG2 and Jiang IFNB/K562, with respectively 60, 60 and 58 retained training interventions. Source feature panels preserve measured genes, and target development/test intervention identities are excluded from source training. Proteomic sources are HCC1143, HCC1806 and MDA-MB-453-1 from mtPTDS, each with 47 retained drug conditions and 3,501 proteins shared with the target panel. They come from one study, not three independent laboratories. Target development/test compounds are excluded from source training. DTI auxiliaries are BioSNAP, Davis and Human from the pinned TAPB release, with 19,092, 1,721 and 2,929 retained training pairs after drug–protein-pair de-duplication against target splits and preceding sources.

The external cell collections are attributed to their original studies (Replogle et al., 2022; Nadig et al., 2025; Jiang et al., 2025a). Proteomic contexts and efficacy labels come from the ProteinTalks supplementary mtPTDS collection (Sun et al., 2026). The source manifest records each retained subset, feature intersection and target-identity exclusion.

Research-model and proposal-budget comparison. Both proposers receive the same 36 configurations: three learning rates, three weight decays, two server-momentum values and two residual-

Target	Source	n	Score	Change	95% CI
VCC	Target only	3/3	29.49 $\pm$ 0.00	0.00 $\pm$ 0.00	[0.00, 0.00]
VCC	+ Replogle	3/3	27.12 $\pm$ 0.08	-2.37 $\pm$ 0.08	[-10.09, 4.26]
VCC	+ Nadig	3/3	26.94 $\pm$ 0.18	-2.55 $\pm$ 0.18	[-11.45, 5.10]
VCC	+ Jiang	3/3	33.07 $\pm$ 0.00	3.58 $\pm$ 0.00	[-2.11, 9.51]
VCC	+ All sources	3/3	27.33 $\pm$ 0.00	-2.15 $\pm$ 0.00	[-12.12, 6.09]
PTPC	Target only	3/3	34.59 $\pm$ 5.66	0.00 $\pm$ 0.00	[0.00, 0.00]
PTPC	+ HCC1143	3/3	34.94 $\pm$ 5.15	0.35 $\pm$ 0.56	[-1.22, 1.60]
PTPC	+ HCC1806	3/3	35.23 $\pm$ 5.35	0.64 $\pm$ 0.32	[-0.78, 1.77]
PTPC	+ MDA-MB-453-1	3/3	35.20 $\pm$ 5.31	0.61 $\pm$ 0.44	[-0.85, 2.04]
PTPC	+ All sources	3/3	35.47 $\pm$ 6.36	0.87 $\pm$ 8.89	[-2.44, 3.99]
DTI	Source comparisons	0/3	N/A	N/A	N/A

Table B.2: External-source transfer on a fixed target test set. Change is paired against target-only training. Scores and changes are mean $\pm$ sample SD, in percentage points. Confidence intervals resample target interventions (VCC) or compounds (PTPC), retaining all paired seed predictions. VCC sources are independent studies; PTPC sources are contexts within the same mtPTDS study.

head settings. Every candidate receives 100 rounds and restarts from the common task/seed initialization. Direct search receives untried identifiers without scores; the loop also receives aggregate development history. Within a backend the first proposal is shared. The proposal budget is 24 slots, not 24 guaranteed valid designs. Each slot allows two attempts to produce the four-field schema. Failed slots remain in the trace. Prefixes at 0, 1, 2, 4, 6, 8, 12, 16, 20 and 24 slots are fixed in advance; selected checkpoints are tested after the full trajectory completes. Configurations evaluated in more than one comparison may share an exact fit cache while retaining their logical budget charge.

Qwen uses the local sampling settings in Appendix A. The second proposer is the gpt-5.6-luna service, accessed in a text-only session with tool use disabled; request metadata and responses are retained. Its service-level sampling controls and parameter count are not disclosed by this interface. The comparison therefore tests proposer backends under common task messages, candidate space and training budget, rather than holding every decoding parameter equal.

Task	Proposer	n	Fixed	Direct@6	Loop@6	Direct@24	Loop@24
DTI	Qwen	0/3	N/A	N/A	N/A	N/A	N/A
DTI	Luna	0/3	N/A	N/A	N/A	N/A	N/A
PTPC	Qwen	1/3	37.27	37.27	37.27	37.27	37.27
PTPC	Luna	1/3	37.27	37.27	37.27	37.27	37.27
VCC	Qwen	1/3	31.50	31.50	31.50	19.61	19.61
VCC	Luna	1/3	31.50	16.94	16.94	20.23	20.01
Norman	Qwen	1/3	15.48	22.30	22.30	22.30	22.30
Norman	Luna	1/3	15.48	35.48	23.11	23.11	23.11
Tahoe	Qwen	1/3	25.37	26.87	26.87	13.43	13.43
Tahoe	Luna	1/3	25.37	13.43	26.87	13.43	13.43

Table B.3: Proposer and search-budget sensitivity. Budgets count proposal slots, including failed proposals. Each included run has completed all 24 slots and evaluation at fixed prefixes. Scores use the original task metric times 100; a single completed seed has no estimated SD. Missing results remain N/A. This larger design-space study is separate from the six-slot main comparison.

The completed seed-61 trajectories also separate proposal execution from search quality. Luna produces valid designs in all 192 attempted slots across the four completed endpoints, whereas Qwen produces 116 valid designs in 192 slots. Accordingly, a flat incumbent curve is interpreted together with the number of executable candidates evaluated. On Tahoe, Qwen’s six-slot direct and loop checkpoints both score 26.87% test Top-1, but both reach 13.43% at 24 slots; Luna also finishes at 13.43% in both modes. Together with the VCC curves, this distinguishes saturation of development selection from improvement on the target task.

Task	Proposer	Direct valid/slots	Loop valid/slots
DTI	Qwen	N/A	N/A
DTI	Luna	N/A	N/A
PTPC	Qwen	19/24	14/24
PTPC	Luna	24/24	24/24
VCC	Qwen	11/24	16/24
VCC	Luna	24/24	24/24
Norman	Qwen	15/24	12/24
Norman	Luna	24/24	24/24
Tahoe	Qwen	14/24	15/24
Tahoe	Luna	24/24	24/24

Table B.4: Valid executable proposals among attempted slots in the completed runs of Table B.3. Each slot allows the same two schema-validation attempts. A larger slot budget need not yield an equal number of trained candidates across proposers.

Search dynamics and task-dependent saturation. Table B.5 compares the slots required to reach a common development target within each pair. Luna reaches Norman’s shared target earlier with feedback (4 versus 22 slots), whereas direct search reaches Tahoe’s shared target earlier (5 versus 13). Proteomics retains the initial design in both modes. Across the 16 completed mode trajectories, the primary score is unchanged in the final eight slots in 12 cases; the full primary-score/loss selection key is unchanged in 10. All trajectories retain between 11 and 24 unvisited configurations. These diagnostics describe finite-budget saturation and identify where additional valid proposals still change selection. The eight paired 24-slot test endpoints give seven ties and one decrease with feedback (VCC/Luna, -0.22 points), so the observed Norman search-efficiency gain does not establish a cross-task final-score advantage.

Task	Proposer	Seed	Dev. target	First slot D/L	Valid D/L	Unvisited D/L
PTPC	Luna	61	66.72	0/0	24/24	11/11
PTPC	Qwen	61	66.72	0/0	19/14	16/21
VCC	Luna	61	30.93	−/10	24/24	11/11
VCC	Qwen	61	24.83	18/23	11/16	24/19
Norman	Luna	61	52.10	22/4	24/24	11/11
Norman	Qwen	61	51.70	2/5	15/12	20/23
Tahoe	Luna	61	35.00	5/13	24/24	11/11
Tahoe	Qwen	61	35.00	18/14	14/15	21/20

Table B.5: Search dynamics at 24 proposal slots. D/L denotes direct/loop. The development target is the higher terminal primary score in each paired comparison; first slot records its earliest attainment, and − means not attained. This target is a retrospective diagnostic, not a stopping rule. Slot 0 is the initial design; valid counts exclude it. Unvisited counts are relative to all 36 configurations, including the initial design. The table covers every completed trajectory in the published snapshot.

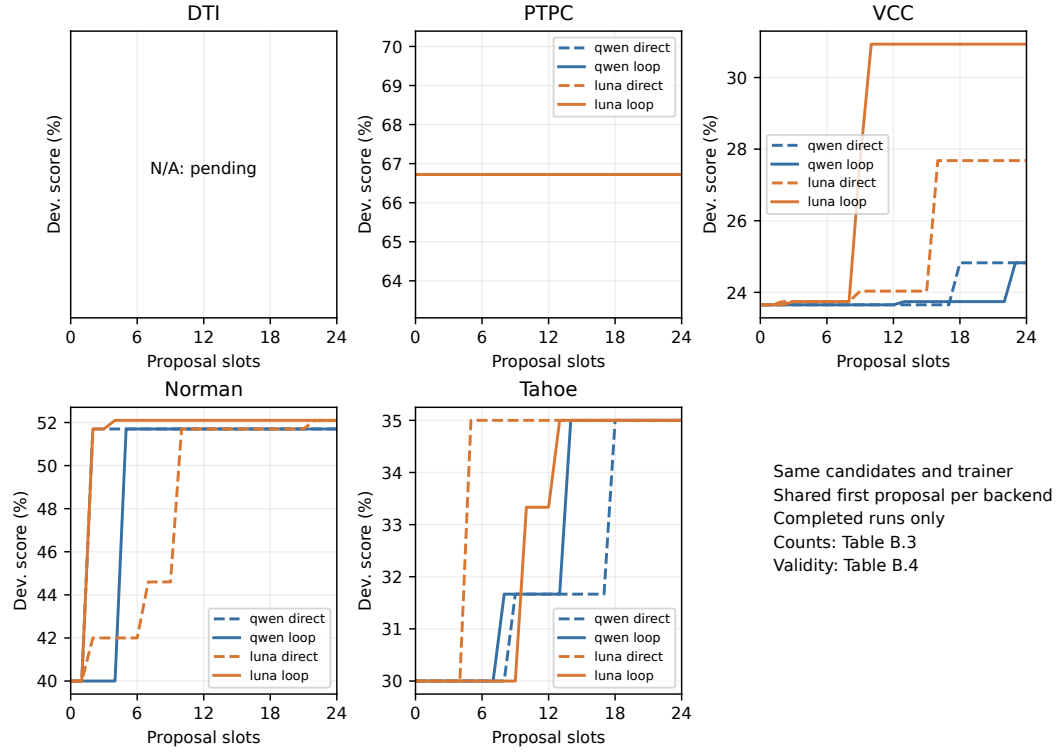


Figure 3: Development-selected incumbent against proposal budget. Only complete trajectories from Table B.3 contribute. A flat primary-score curve may still contain a lower-loss checkpoint selection; failed proposal slots also remain on the horizontal axis.

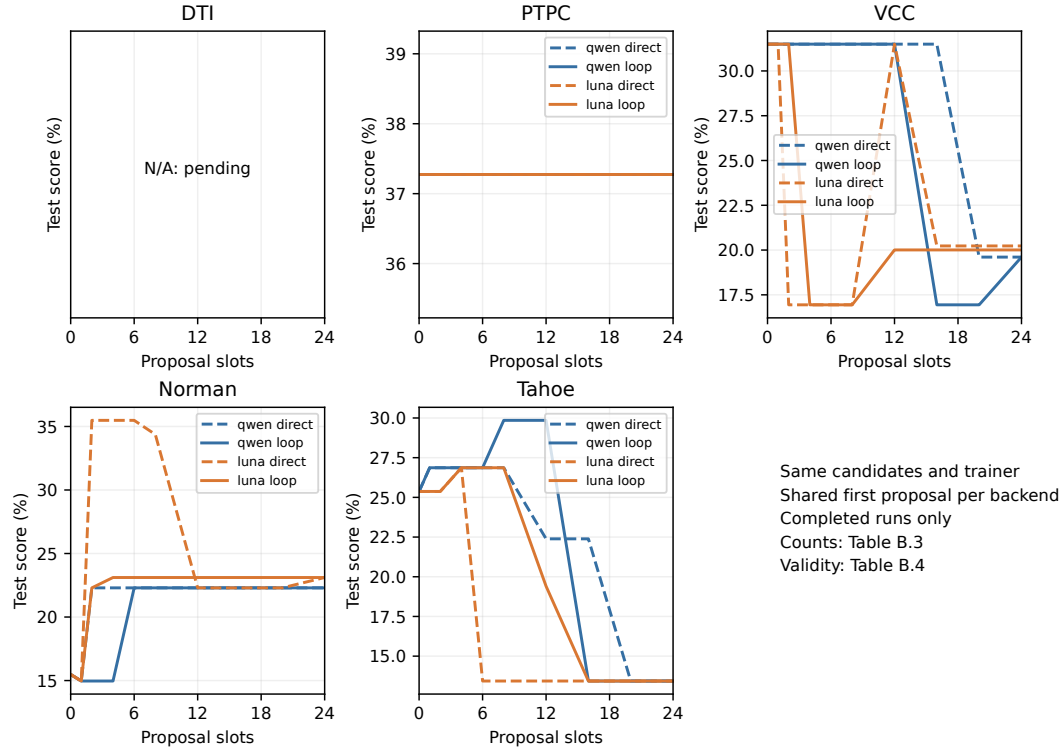


Figure 4: Held-out scores of the development-selected checkpoints at fixed proposal prefixes. Improved development search need not improve the target test metric. The full prefix series is shown, rather than choosing a budget from its highest test score.